

AEGIS STUD - Assignment Workflow Engine

Product, UX, orchestration, academic-quality and local-compute specification

Draft specification for continued AegisUi STUD development

Scope target: Master's-level submission-ready candidate, with mandatory human review before submission.

Core design principle: automation without opacity. Every important source, claim, decision, milestone, blocker, revision and output must remain inspectable and correctable by the student.

1. Executive intent

STUD is not to be designed as a single 'write my assignment' button. It is an academic workflow operating system that can ingest an assignment from an LMS or manual input, reconstruct its requirements, plan and execute a bounded research-and-writing workflow, expose progress in real time, preserve provenance, stop honestly at external dependencies, and produce a high-quality final candidate document for student review.

- Primary quality target: a coherent, evidence-grounded, citation-verified candidate document capable of meeting Master's-level expectations when the available brief, evidence, model capability and subject matter permit it.
- Primary UX target: Apple-like simplicity at rest; sci-fi mission-control visibility only while complex work is actually running.
- Primary trust target: no fabricated activity, sources, citations, experimental data, team contributions or hidden automatic submission.
- Primary architecture target: discipline-neutral. Engineering is a supported discipline, not the canonical model.
- Primary deployment target: local-first and offline-capable after required materials/models are present, with explicit use of external academic providers only where required.

2. Assignment ingress: Moodle is optional, STUD is canonical

The canonical Assignment object belongs to STUD. Moodle or any future LMS is only a source of observations and managed artifacts. The workflow must therefore remain fully usable in universities, colleges and schools where LMS scraping/API access is unavailable, incomplete, prohibited, or simply not used.

Ingress route	Expected behaviour
Moodle/LMS	Discover courses and assignments; import allowed metadata/files; retain source provenance; never make STUD structurally dependent on Moodle.
Manual assignment	First-class creation route. User can enter module, title, deadline, word count, assessment type and free-form instructions.
Imported brief	PDF/DOCX/image/scan or other supported material is attached to an Assignment and analysed into proposed requirements.
Quick capture	Minimal text or dictated description creates an incomplete Assignment immediately; missing fields remain UNKNOWN and can be enriched later.
Paper/verbal instructions	User photographs, scans, types or dictates the instructions. STUD treats these as evidence, not as

	complete truth until reviewed.
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Architectural rule: Moodle is a provider of context; it is never the owner of the Assignment.

3. Canonical assignment intake and requirements reconstruction

3.1 Intake

1. Create or discover the Assignment.
2. Collect all known briefs, module handbooks, rubrics, learning outcomes, lecturer notes, LMS metadata and submission instructions.
3. Normalize dates/times/timezones and preserve original observations.
4. Extract candidate requirements without silently resolving contradictions.
5. Present a human-reviewable Requirements Contract before high-cost execution.

3.2 Requirements Contract

Field	Examples / behaviour
Assessment identity	Course/module code, assessment name, component number, weighting.
Deliverable	Essay, report, portfolio, presentation, code, notebook, lab report, group work, dissertation component, other.
Academic level	Foundation / undergraduate levels / Master's / custom institutional level.
Length	Word/page/time limit; inclusion/exclusion rules; tolerance if explicitly stated.
Deadline	Release, due, cutoff, resubmission where known.
Learning outcomes	Mapped exactly to source wording where available.
Rubric	Criteria, bands, descriptors, weightings, pass conditions.
Evidence constraints	Minimum/expected sources, date ranges, source types, prescribed reading.
Formatting	Harvard/APA/Vancouver/other, structure, appendices, figures, naming, file type.
Integrity/policy	Institutional or assignment-specific permitted-use constraints stored as requirements, not generic moral assumptions.
Unknowns/conflicts	Explicitly surfaced; never guessed.

4. Workflow model: milestones on screen, dependency graph underneath

The user sees a comprehensible milestone timeline. Internally, the workflow is a dependency DAG so independent branches can proceed in parallel and an external blocker does not freeze unrelated work.

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INTAKE -> REQUIREMENTS -> RESEARCH PLAN -> SOURCE ACQUISITION
                                     -> FACULTY LITERATURE SCOUT
SOURCES -> TOPIC DOSSIERS -> EVIDENCE MAP -> OUTLINE -> DRAFT
EXTERNAL DATA -----> RESULTS/DISCUSSION
DRAFT -> INTEGRATION -> CITATION PASS -> HUMANISATION -> COMMITTEE
COMMITTEE -> CORRECTIONS LOOP -> FINAL PACKAGE -> HUMAN REVIEW

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4.1 Proposed milestone contracts

Milestone	Required output / gate
M0 Intake	Canonical Assignment exists; source provenance retained.
M1 Requirements	Reviewed Requirements Contract; unresolved fields explicit.
M2 Research Plan	Questions/topics, search plan, evidence needs, stopping criteria.
M3 Source Acquisition	Verified candidate library; quality/relevance screening; faculty scan.
M4 Topic Dossiers	Per-topic evidence, counterevidence, gaps, source provenance.
M5 Evidence Map	Claims/evidence/requirements matrix; unsupported claims visible.
M6 Composition Plan	Outline, argument flow, section purpose, word budget.
M7 Draft	Section drafts grounded in evidence; no invented external results.
M8 Integration	Cross-section coherence, transitions, duplication removal, argument consistency.
M9 Citation Integrity	Citation formatting plus claim-support verification.
M10 Humanisation	Voice adaptation without deliberate errors or deceptive source alteration.
M11 Committee	Rubric-based multi-marker critique and predicted band/range, explicitly non-authoritative.
M12 Corrections	Issues triaged and resolved through versioned revisions.
M13 Final Package	Submission-ready candidate + bibliography/appendices/required artifacts/checklist.
M14 Student Review	Mandatory final human inspection; no automatic submission.

5. External blockers and honest incompleteness

Assignments frequently depend on events that software cannot legitimately automate: laboratory sessions, fieldwork, interviews, surveys, supervisor feedback, approvals, group members or externally supplied datasets. These dependencies are first-class workflow objects.

Status	Meaning
READY	Dependencies satisfied; may execute.
RUNNING	Actively executing real work.
COMPLETE	Milestone contract satisfied.
PARTIAL	Useful work completed but requirements remain.
BLOCKED	External dependency prevents completion.
NEEDS_REVIEW	Human decision required.
FAILED	Execution error; not equivalent to blocked.
CANCELLED	Stopped by user/system policy.

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5.1 Blocker taxonomy

- WAITING_LAB
- WAITING_TEAM_MEMBER
- WAITING_DATA
- WAITING_FEEDBACK
- WAITING_SUPERVISOR
- WAITING_APPROVAL
- WAITING_RESOURCE
- WAITING_EVENT
- WAITING_INTERVIEW
- WAITING_SURVEY
- WAITING_FIELDWORK
- CUSTOM

A blocker records what is missing, why it is required, owner, expected date if known, required artifact, downstream dependencies and which branches may continue.

Hard rule: an external-world blocker may never be resolved by generating fictional data or fictional human contributions.

5.2 Specialized laboratory behaviour

- Pre-lab: parse procedure, variables, expected measurements, equations, hazards if supplied, blank data tables and analysis plan.
- During wait: complete independent background, literature and methodology framing.
- Post-lab: ingest user-supplied measurements, validate structure, perform permitted calculations/plots, then unlock results/discussion branches.
- If data are absent, Results remains BLOCKED; the system may create placeholders describing what evidence is required but not fabricate observations.

6. Research engine

6.1 Research plan

Research begins only after the Requirements Contract is sufficiently understood. The plan decomposes the assignment into research questions/topics and defines evidence requirements per topic.

6.2 Source quality gate

- Topical relevance and direct contribution to a defined question.
- Authority and publication type.
- Peer-review status where relevant and knowable.
- Currency where the topic is time-sensitive.
- Primary vs secondary evidence distinction.
- Duplicate/derivative detection.
- Methodological relevance and limitations.
- Contradictory or dissenting literature deliberately sought.
- Source function: background, evidence, counterargument, method, definition, case, data, theory, etc.

6.3 Research stopping criterion

The agent must not search indefinitely. A topic may reach saturation when its required claims have adequate high-quality support, meaningful counterevidence has been considered, critical evidence gaps are resolved or explicitly blocked, diversity thresholds are met, and further searches produce diminishing relevant novelty.

6.4 Faculty Literature Scout - GEM feature

Teaching-team names found in official university/module materials may be resolved against scholarly identities. Relevant publications may be surfaced as rare 'Faculty Gems' without being artificially promoted over better evidence.

Identity evidence	Use
Institutional name/profile	Initial teaching-team identity.
Affiliation/department	Disambiguation.
ORCID	High-confidence scholarly identity where available.
OpenAlex/Crossref metadata	Publication discovery and metadata corroboration.
Coauthors/topics/dates	Secondary disambiguation.

- States: CONFIRMED / PROBABLE / AMBIGUOUS.
- A faculty publication must pass the same relevance/quality gate as any other source.
- Do not force inclusion and do not use flattering language in the final document.
- Surface at Topic Dossier and Evidence Map level only when genuinely relevant.

7. Evidence, claims and provenance

Target traceability: Draft sentence -> Claim -> Evidence excerpt -> Source -> page/chunk/document; and reverse navigation from source to every downstream use.

- Every material claim should be traceable to evidence, explicit reasoning, user-authored observation, or a clearly marked unsupported/gap state.
- Evidence excerpts retain document identity, page/chunk, extraction method and hashes where appropriate.
- Figures, tables, equations, datasets and code receive provenance equivalent to text.
- Citation integrity is distinct from citation formatting: a real DOI does not prove that the cited work supports the sentence.
- Contradictory evidence must remain visible rather than being silently discarded.

8. Composition engine

8.1 Outline and argument design

- Outline derived from requirements and evidence rather than generic essay templates.
- Each section records purpose, requirements/LOs served, planned claims, evidence and target word budget.
- The engine performs a pre-mortem before drafting: 'If this scored poorly, what structural/evidential reason would explain it?'
- The pre-mortem may force revision of the outline before expensive drafting begins.

8.2 Continuous word-budget controller

Introduction	278 / 300	
Literature review	1042 / 900	OVER
Discussion	611 / 1100	UNDERDEVELOPED
Conclusion	0 / 250	PENDING

8.3 Requirement coverage

Coverage is live throughout drafting, not a final checklist. Learning outcomes, rubric criteria and explicit brief requirements map to evidence and draft sections. UNKNOWN or AT RISK must remain visible.

8.4 Master's-level quality target

- Critical synthesis rather than serial summary.

- Explicit engagement with limitations, counterarguments and methodological quality.
- Argument coherence across sections.
- Evidence density appropriate to the discipline and assignment.
- Accurate academic register without generic AI filler.
- No fabricated facts, quotations, citations or external results.
- Final output target is submission-ready candidate quality, not intentionally mediocre 'draft quality'.

9. Humanisation / author voice

Humanisation is a controlled editorial stage, not an instruction to inject spelling mistakes or evade detectors. Its purpose is to reduce generic machine prose and align the candidate document with a natural, consistent authorial voice while preserving evidence and meaning.

- Support multiple writing fingerprints by genre: essay, lab report, technical report, reflective writing, etc.
- Prefer user-provided historical samples or explicitly configured preferences.
- Version the transformation and expose a diff: composition -> humanisation -> final author edits.
- Never alter citations, evidence meaning or numerical results merely to sound more human.
- Flag suspiciously formulaic repetition, empty transitions, excessive hedging, inflated vocabulary and generic conclusions.

10. Lecturer Committee and quality-control loop

The final evaluation stage simulates a panel of markers at the appropriate academic level. Where an official rubric exists, it is the primary scoring framework. The committee's grade is an estimate, not an institutional prediction.

Panel role	Focus
Primary marker	Rubric compliance, academic quality, coherence.
Second marker	Independent reading and disagreement detection.
Methods/evidence reviewer	Evidence quality, methodology, figures/data and claim support.
External examiner / devil's advocate	Weak assumptions, missing counterarguments, overclaiming, structural risk.
Citation reviewer	Reference accuracy, support integrity, style consistency.

- Committee cannot penalize a section as 'poorly completed' when it is explicitly BLOCKED by a genuine external dependency; it must report incomplete status separately.
- Corrections are versioned and linked to committee findings.
- Committee rounds may repeat until quality gates are satisfied or the user stops.
- After real marking, lecturer grade/feedback can be imported and compared with committee findings to improve future evaluation heuristics without rewriting historical results.

11. Assignment Run - Mission Control

High-cost automated execution is launched manually. The student chooses when the Mac may be heavily occupied. Launch is preceded by an explicit resource and capability summary.

11.1 Launch panel

COURSEWORK 01	
3,500 words Due 22 Nov 14:00 Weight 40%	
QUALITY TARGET	MASTER'S
RESOURCE PROFILE	MAXIMUM / LEAVE MAC ALONE
LOCAL AI	READY
AEGIS STUD Assignment Workflow Engine Specification - development draft	

EXTERNAL SOURCES REQUIRED
KNOWN BLOCKERS 0

This run may use substantial CPU, RAM, storage and network bandwidth.

[CANCEL] [LAUNCH ASSIGNMENT RUN]

11.2 Mission Control layout

Top: SpaceX-style milestone strip. Main body: two-pane operational view. Left pane is an Artifact Bay showing real acquired/created objects. Right pane is a real-time preview of the operation actually being performed. Bottom/secondary area is a concise operational activity feed and system telemetry.

Area	Contents
Milestone strip	Completed/current/pending/partial/blocked/review states; clickable history.
Artifact Bay	PDFs, URLs, DOI records, citations, images, datasets, notes, excerpts, dossiers, drafts, figures.
Live Work	Current PDF/page, search result set, evidence comparison, outline section, draft revision or committee review.
Operational feed	Timestamped meaningful events; not developer logs.
Telemetry	CPU, memory, disk I/O, network, current task runtime, last progress.
Controls	Pause, resume, give Mac back, retry, skip where safe, cancel, inspect artifact.

No fake cinematic telemetry. If Mission Control says 'reading page 17', the system must actually be processing that page or a derived object from it.

11.3 Watchdog and stalls

- Track last meaningful progress separately from elapsed wall-clock time.
- If a task exceeds its expected bound without progress, show POSSIBLE STALL rather than pretending activity.
- Offer WAIT, RETRY CURRENT TASK, SKIP (only if dependency rules allow), PAUSE RUN and CANCEL.
- Failures should degrade locally: an unavailable provider should not kill independent branches.

11.4 Resource profiles

Profile	Intent
ECO	Background execution with conservative concurrency and memory.
BALANCED	Default tradeoff.
MAXIMUM	User intentionally leaves the Mac to STUD; maximize safe local throughput.
CUSTOM	Explicit limits for RAM, concurrency, network and other supported resources.

A 'GIVE MAC BACK' action should checkpoint safely, reduce/stop workers and return resources without corrupting the run.

12. Checkpointing, recovery and reproducibility

- Checkpoint at every milestone and at bounded substeps within long milestones.

- Persist enough orchestration state to resume after app crash/restart without replaying completed expensive work.
- Each run receives a stable RUN ID and immutable execution metadata.
- Record model/profile, relevant settings, source set, context package IDs, milestone outputs, errors, blockers and versions.
- Allow inspection/rollback to earlier candidate versions without deleting provenance.
- Never claim bit-for-bit reproducibility for stochastic model output unless the runtime/configuration actually supports it.

13. Model architecture and expected capability

The workflow must be model-agnostic. Model size is a capability/resource choice, not part of the canonical academic schema.

Tier	Typical work
Small / fast (~3B-4B)	Classification, metadata, requirement extraction assistance, tagging, triage, routing, simple summaries.
Medium (~8B)	Intermediate synthesis and dossier support where quality permits.
Academic worker (~12B-14B on 16 GB class hardware)	Topic synthesis, outline reasoning, section drafting, integration, humanisation and committee work, accepting slower execution.
High-end (~27B-32B+)	Higher quality ceiling on hardware with sufficient unified/system memory.
Very large (~70B+)	Optional future high-end local deployment; never required by workflow design.

For a 16 GB Apple Silicon Mac, a quantized 12B-14B class academic worker is the practical initial target. The 3B model remains useful as a fast worker but should not define the Master's-level quality ceiling.

Long runtime is acceptable by design: the intended usage includes overnight or otherwise unattended runs. Quality and inspectability take priority over speed.

14. Storage architecture - internal SSD + external academic volume

Large models, managed Moodle material, PDFs, datasets, run artifacts and caches may live on a user-selected external HDD/SSD. Critical STUD metadata should remain lightweight and locally accessible so the workspace can still show assignments, deadlines and state when the external volume is absent.

Internal Mac SSD

AegisUi
STUD SQLite / critical metadata
lightweight configuration
small active assets

User-selected Academic Storage (e.g. external Seagate)

Models/
Moodle/
AcademicLibrary/
AssignmentRuns/
Datasets/
ManagedDocuments/
Cache/

14.1 Storage abstraction

- Never hardcode a user's volume name or path in product logic.
- Store a stable storage profile/volume identity and managed relative references.
- Detect disconnected storage and mark dependent assets OFFLINE rather than corrupting or silently duplicating them.
- Provide usage totals and safe cache cleanup.
- Classify artifacts as canonical, referenced, temporary or obsolete to support garbage collection.

14.2 Portable assignment mode

Before travel, the user may request MAKE ASSIGNMENT PORTABLE. STUD computes the minimal local package: required PDFs/datasets/notes/context plus, optionally, the chosen local model. It copies these to internal storage with explicit size requirements, then can return them to external storage later.

14.3 HDD expectations

An HDD is acceptable when runtime latency is not a priority. Initial model loading and many-small-file operations will be slower than on SSD, but the design intentionally supports long unattended runs. An external SSD is an optimization, not a prerequisite.

15. UX direction for the wider STUD module

STUD should move away from a flat wall of blue rectangles and toward contextual workspaces. Advanced capabilities remain available but are progressively disclosed.

- Primary navigation centered on Courses, Assignments and the current work context.
- Contextual tabs/menus rather than fourteen equally prominent surfaces.
- Persistent preview area for PDF, document, GitHub, web/video references or other active source types where technically supported.
- Adjacent notes area that can receive generated notes and direct user edits.
- Current Assignment/Context is inferred from explicit navigation and relationships, with a visible correction control when inference is wrong.
- Moving to Citations, Knowledge, Notes or Research should carry the active Assignment context automatically rather than forcing re-selection.
- Knowledge should surface relevant adjacent/alternative material specifically connected to the current line of work, not generic recommendations.
- At rest, hide instrumentation that is not actionable. During Assignment Run, reveal Mission Control.

16. Security, privacy and user control

- Persist authorized LMS configuration/identity and encrypted credentials when the user explicitly configures an account; persistence is necessary for useful synchronization. Provide a clear Forget Account action.
- Do not confuse secure persistence with non-persistence: credentials may be retained securely while authenticated transient URLs and raw secrets are excluded from academic records/logs.
- No generic renderer-controlled HTTP proxy, arbitrary shell or arbitrary filesystem execution.
- External provider calls are explicit capabilities with typed boundaries.
- No telemetry requirement for core STUD.
- No hidden automatic submission to Moodle/LMS.
- No fabricated progress UI.
- No silent destructive resolution of provenance conflicts.
- No automatic creation of experimental/team/interview evidence.

17. Output package

- Final candidate document in required format(s) where supported.

- Bibliography/reference list in assignment-required style.
- Appendices, figures, tables, datasets/code references as required.
- Requirement/LO/rubric coverage checklist.
- Citation integrity report.
- Committee report and unresolved issues.
- External blocker report if incomplete.
- Run provenance/audit trail accessible separately from the submitted document.
- Submission remains a user action.

18. Post-assignment learning loop

- Import real grade and lecturer feedback when available.
- Map feedback to rubric criteria, sections and prior committee findings.
- Compare predicted range with actual result without retroactively changing the old prediction.
- Identify committee misses and false alarms.
- Use accumulated local evidence to improve future heuristics/profile choices, with transparent provenance and user control.

19. Acceptance scenarios

Scenario	Pass condition
Moodle essay	Assignment discovered/imported, brief/rubric reconstructed, user reviews and launches run, final candidate plus audit artifacts produced.
Manual paper brief	User creates Assignment and attaches scan/PDF; workflow is equivalent after intake.
Verbal assignment	Quick Capture creates partial Assignment; later evidence enriches it without data loss.
Lab report	Independent work proceeds; results/discussion remain BLOCKED until real dataset arrives.
Group project	User/team ownership explicit; missing teammate artifacts block only dependent branches.
Offline flight	Portable assignment package and local model permit work without internet; external-source tasks remain deferred.
External HDD disconnected	Assignments/deadlines remain visible; dependent assets/models show OFFLINE and recover on reconnect.
Crash mid-run	App resumes from recent checkpoint rather than restarting entire assignment.
Weak model only	System warns LIMITED MODEL CAPABILITY and does not claim Master's-equivalent quality.
Faculty Gem	Lecturer identity resolved and relevant publication surfaced only after normal quality gate.
Citation audit	User can trace a claim to exact evidence/source and detect unsupported attribution.
Final review	No submission occurs until explicit student action.

20. Development priorities / recommended implementation order

1. Canonical manual Assignment creation and Requirements Contract.
2. Workflow DAG, milestone contracts, blocker objects and checkpoint persistence.
3. Contextual Assignment-centric UX refactor with preview + notes.
4. Mission Control shell, real event telemetry and Artifact Bay.
5. Research plan, source-quality gate, stopping criterion and Topic Dossiers.
6. Claim/evidence/citation integrity graph and requirement coverage.
7. Composition plan, word budget, drafting and integration.
8. Faculty Literature Scout.
9. Humanisation pipeline and author fingerprint profiles.
10. Lecturer Committee, pre-mortem, corrections loop and post-grade feedback.
11. Resource profiles, watchdog, pause/resume/give-Mac-back.
12. External academic storage abstraction, garbage collection and portable assignment mode.
13. End-to-end Master's-level acceptance run using a real or faithfully reconstructed brief and rubric.

21. Non-goals and truth constraints

- Do not promise exact institutional grades.
- Do not claim semantic understanding from deterministic keyword overlap.
- Do not claim Moodle can be fully scraped at every institution; capability is deployment-specific.
- Do not fabricate inaccessible papers or pretend paywalled material was read.
- Do not fabricate lab data, interviews, surveys, team work or feedback.
- Do not use cinematic fake logs to mask inactivity.
- Do not deliberately degrade final output merely because the student is expected to edit it.
- Do not make a particular LLM/model/provider a schema dependency.

22. Definition of success

A successful STUD Assignment Workflow Engine allows a student to move from an imperfect real-world brief to an inspectable, evidence-grounded, high-quality candidate submission through a workflow that is understandable at a glance and auditable in depth. The user can leave the machine working for hours, see exactly what it is doing, resume after interruption, inspect every artifact, correct any stage, and trust that genuine missing real-world inputs remain visibly missing rather than being invented.

Product character: less enterprise dashboard, more calm Apple workflow at rest; more SpaceX/CIA-style operational visibility only when real work is running.